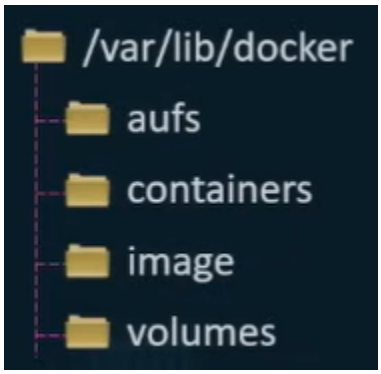
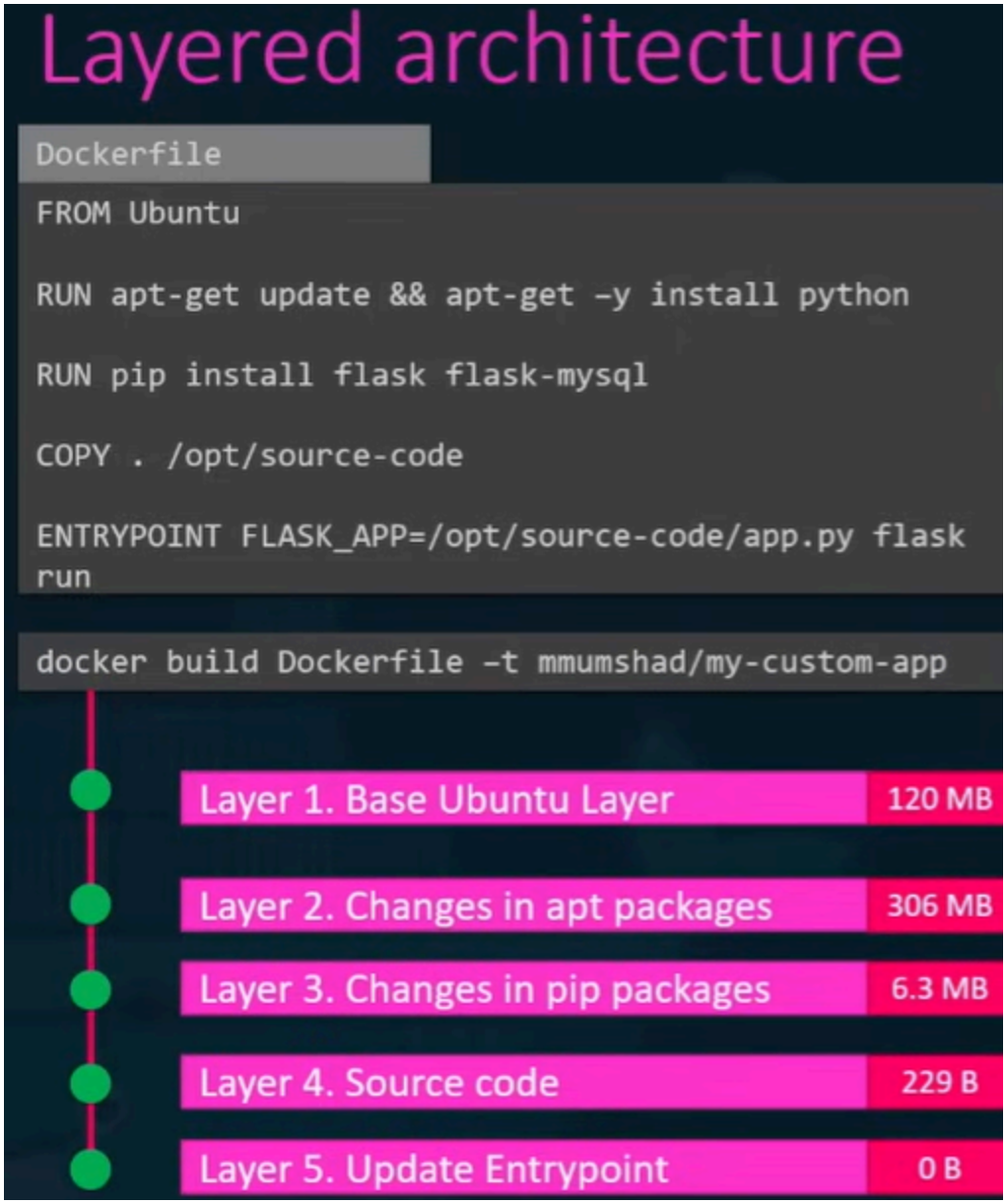
Docker Storage

After installing docker, it creates this folder structure,



where file related to images and data is stored default. Images, containers running in local host. So how does the docker stores the images and containers to corresponding folders? To understand that, let's understand Layered Architecure,

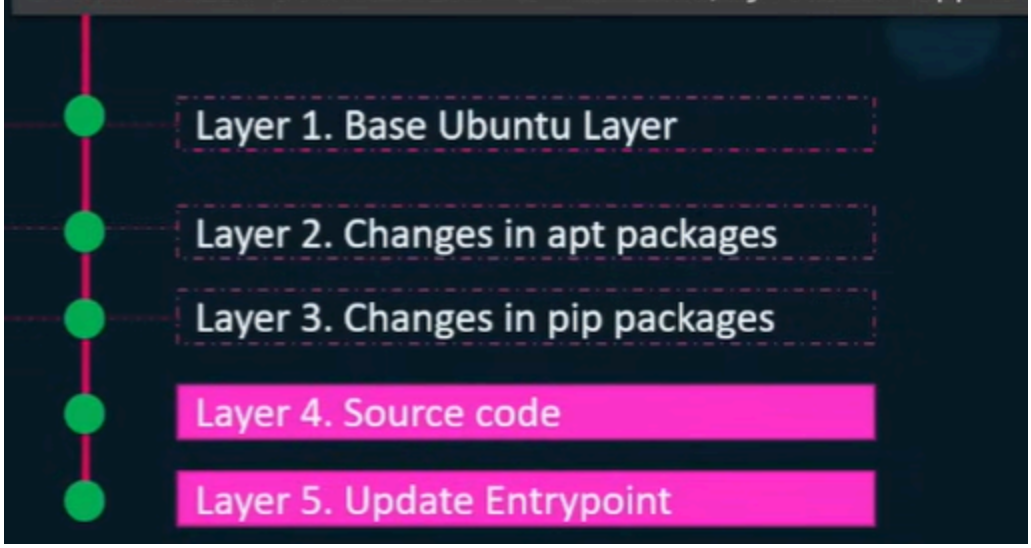
Each line of instruction, in the docker file, creates a new layer in the docker image, just a changes from the previous layer,



Advantage of layered architecture, let’s consider second application,

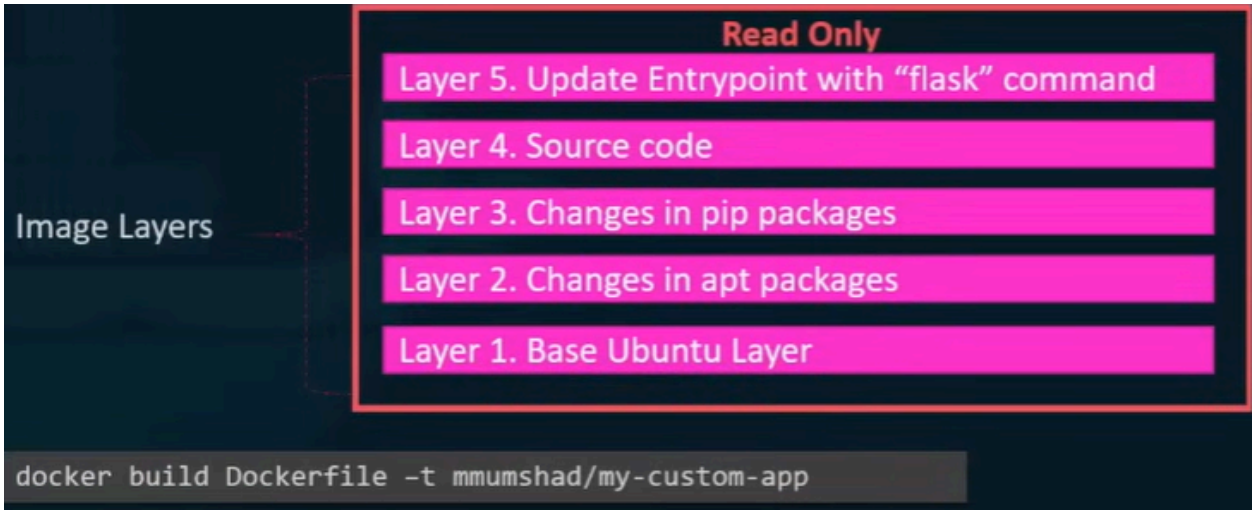
```
Dockerfile2
FROM Ubuntu

RUN apt-get update && apt-get -y install python

RUN pip install flask flask-mysql

COPY app2.py /opt/source-code

ENTRYPOINT FLASK_APP=/opt/source-code/app2.py flask
run
```

```
docker build Dockerfile2 -t mmumshad/my-custom-app-2
```

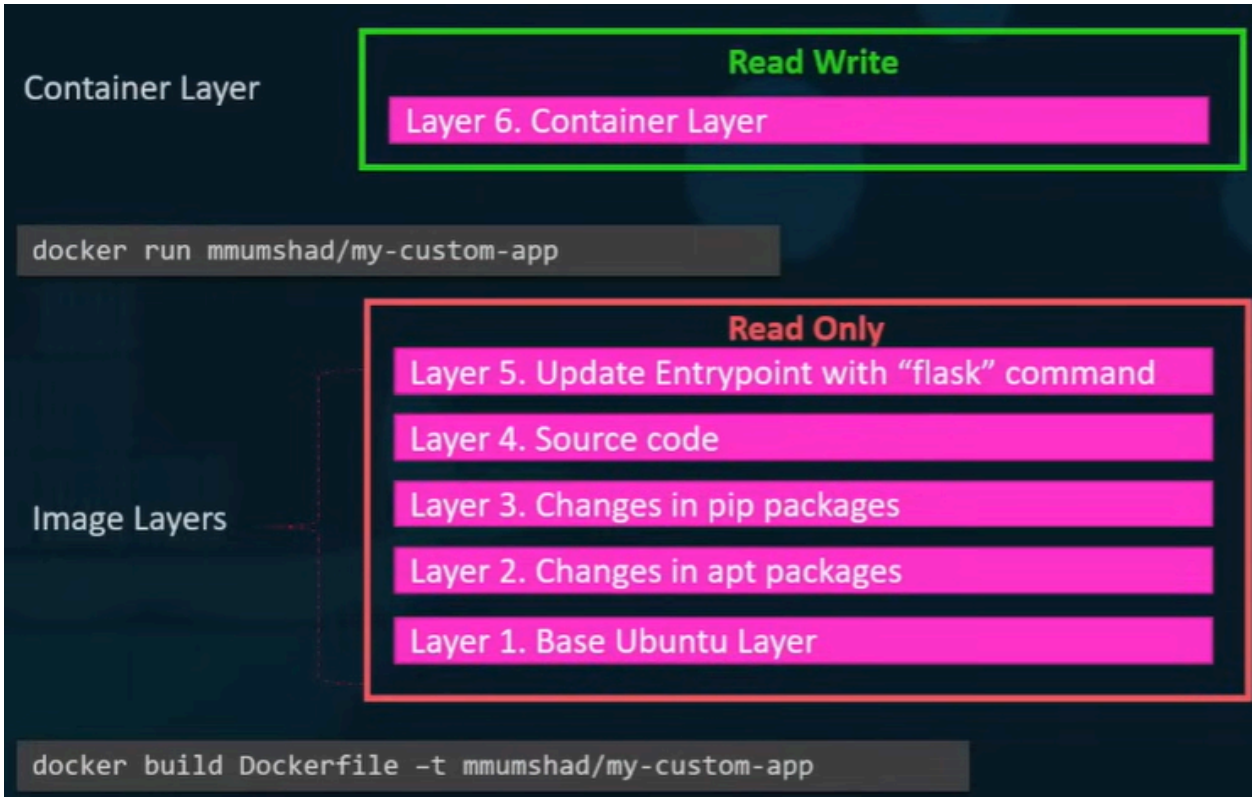


The above application has the different docker file, very similar to first application, very similar to the previous docker file, to create a docker image using built command, since first three layers of both the docker file are the same, now docker does not going to build the first three layers, instead it uses the first three layers of the previous application from the cache, and creates last two layers for source code and entrypoint. This way docker create images faster, efficiently saves disk space. This is also applicable, if you want to update the application code. Whenever, you want to update the application code, docker simply uses previous layers cache, and quickly rebuilds the application image by updating the latest source code. Thus, saving lot of time, clearing rebuilds, and updates.

Re-arranging the layer bottom up,



Once the built is complete, you cannot create the context of this layer, they are read only. You can only modify themby initiating new build.when you run the container, based of this image, using docker run command, create a writable layer above the image layer, writable layer is used to store data created by container, such as log, temporary files, or any file modified by the user,



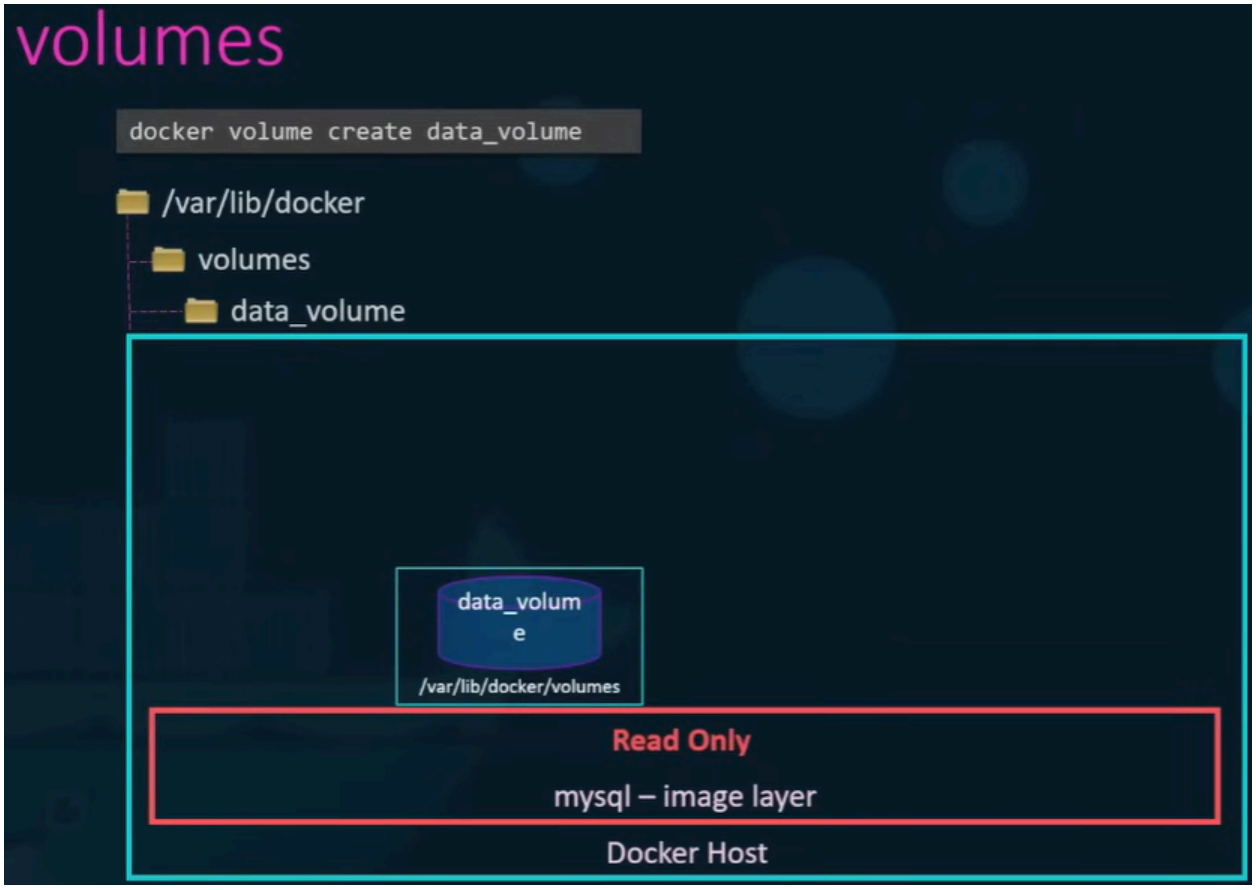
Life of this layer 6: as long as the container is alive.

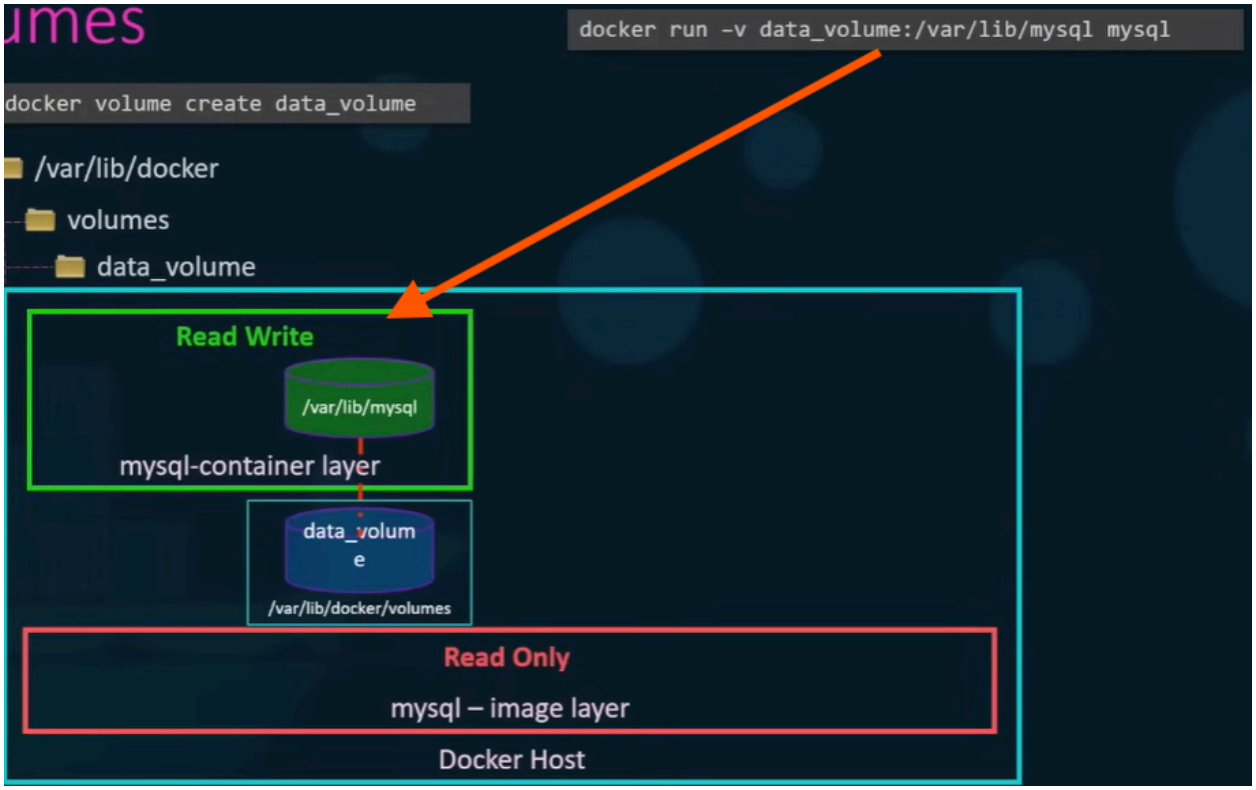
To modify, docker creates copy of that file in read-write,

Known as copy write mechanism,

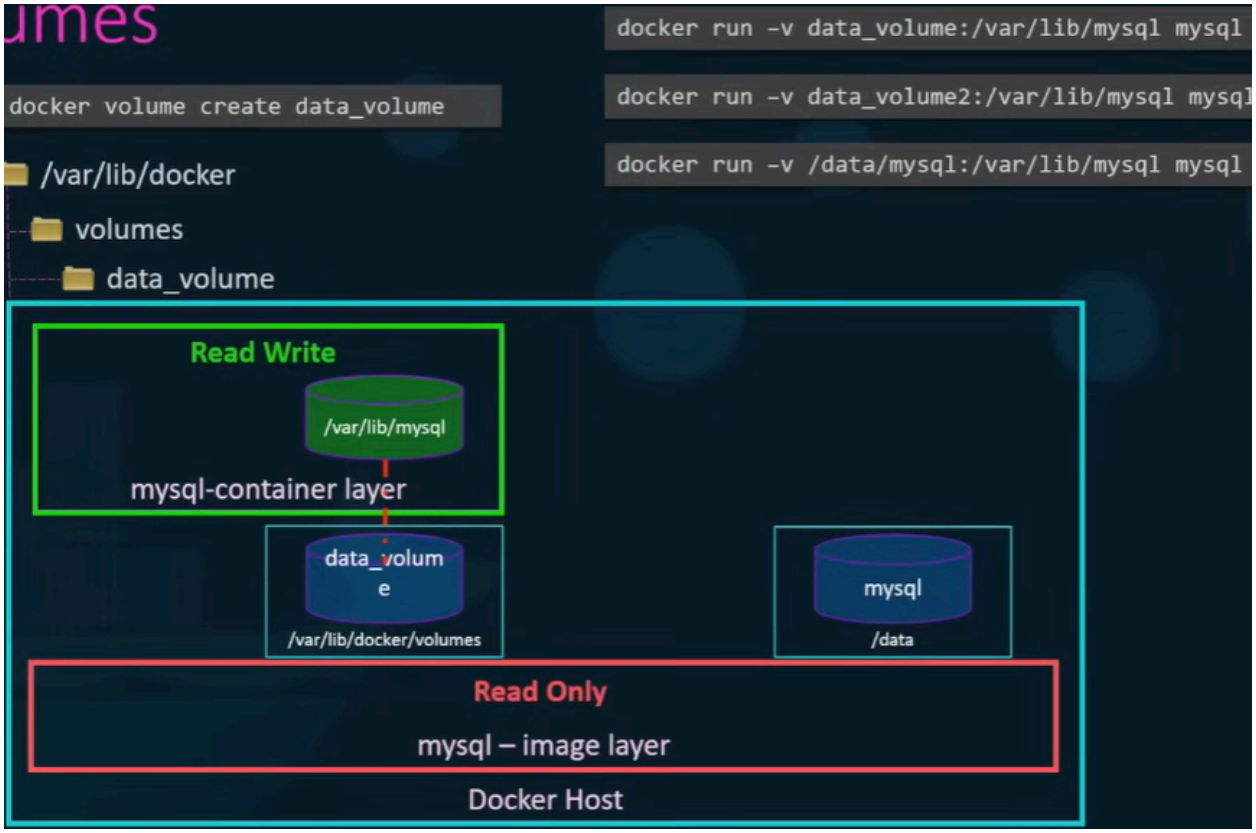
Container Layer
Read Write
app.py
temp.txt
Read Only
Image Layers
app.py

Volumes:

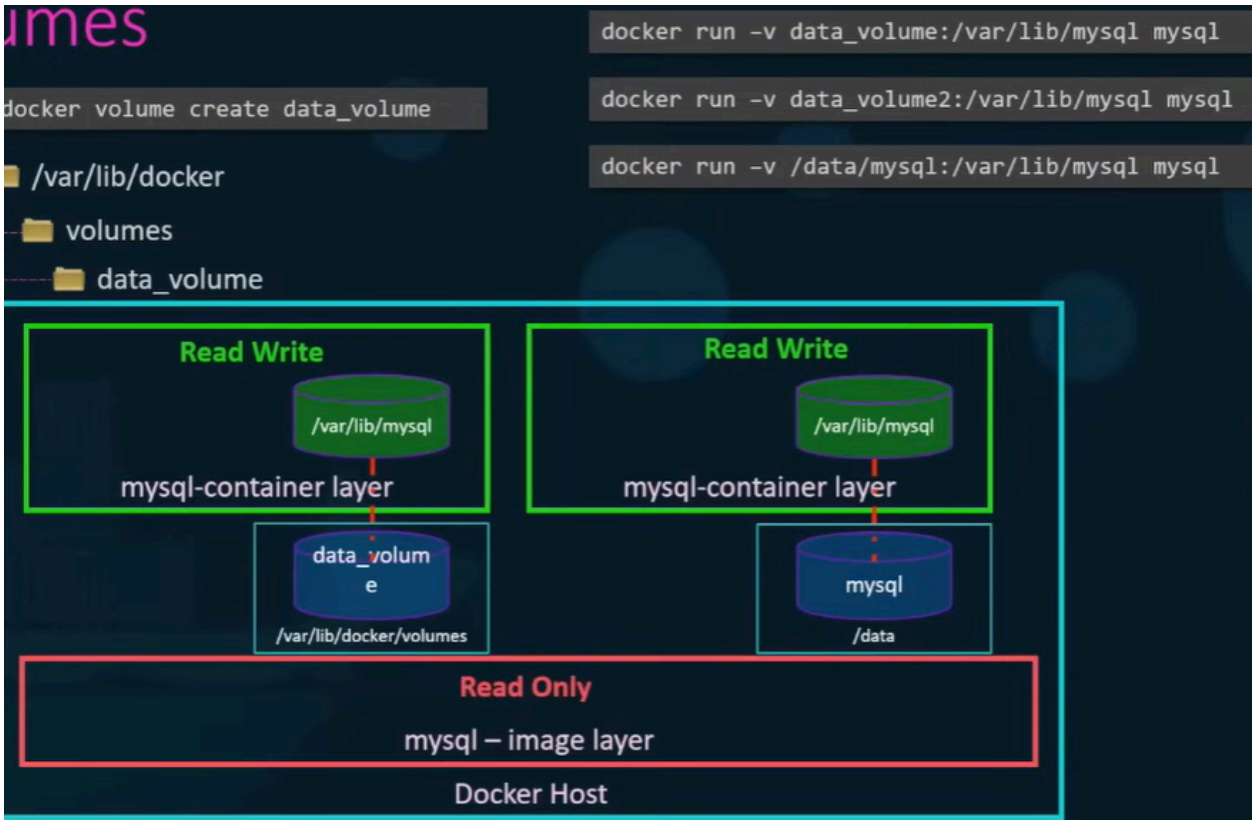

volumes
docker volume create data_volume
/var/lib/docker
volumes
data_volume
data_volum
e
/var/lib/docker/volumes
Read Only
mysql – image layer
Docker Host



Creates a new layer that mounts the new datalayer created(data_volume) into /var/lib/mysql folder inside the container folder,



There are two types of mount,
Volume mounting and bind mounting.



https://youtu.be/fqMOX6JJhGo?list=PLG_szO4yFOP692mrsGEbhSx9Fj3Aa3h_z&t=4292